

Spatial and Seasonal Patterns in Key Water Quality Indicators Across Illinois Monitoring Sites

Team Members: Nolan Rink

Selected Project Topic: Option 2 – Water Quality Data Analysis

Problem Statement

This project investigates how cleaned measurements of dissolved oxygen, pH, phosphorus, and specific conductance vary across Illinois monitoring sites and seasons from 2021 to 2025. These indicators were selected after exploratory data analysis showed they have the strongest combination of numeric usability, site coverage, temporal coverage, and unit consistency in the available Water Quality Portal export. The study is important because spatial and seasonal variation in these core indicators can reveal meaningful differences in monitored water conditions across Illinois while still keeping the project scope realistic and defensible.

Dataset Description

The dataset is an Illinois-focused physical and chemical water quality results export from the Water Quality Portal (WQP), a public data source maintained through cooperation among the U.S. Geological Survey, the Environmental Protection Agency, and the National Water Quality Monitoring Council. The current file contains 85,619 rows and 81 columns, covering 857 observed monitoring sites and 42 distinct characteristics, with parsed date coverage from January 3, 2021 through December 31, 2025.

Key fields include monitoring location identifiers and names, latitude and longitude, activity start dates and times, characteristic names, result values, result units, and quality-related fields. The exploratory analysis also identified 7,669 likely duplicate measurement keys, several sparse columns, and a small number of multi-unit characteristics, which makes preprocessing an essential part of the project design.

Planned Analysis Approach

The analysis will begin by mapping the observed schema to site, date, characteristic, value, and unit roles, then cleaning the dataset by identifying duplicate measurement keys, reconciling units where appropriate, and flagging suspicious extreme values with conservative rules. Exact duplicate measurement keys will be aggregated in cleaned summaries while raw rows are preserved for audit. The main analysis will focus on dissolved oxygen, pH, phosphorus, and specific conductance, while dissolved oxygen saturation will be used as a supporting indicator. Summary statistics, seasonal comparisons, site-level comparisons, visualizations, and selected data mining methods such as clustering or correlation analysis will be used to identify major patterns across the monitored Illinois sites.

Related Work or Existing Results

Prior work using the Water Quality Portal gives a useful foundation for this project. For example, Read et al. (2017) described the Water Quality Portal as a national-scale resource for aquatic research, and Mohebbi and Lee (2024) used WQP-based river and stream data to study long-term nitrate and nitrite trends across the contiguous United States. In Illinois, this project also fits with how the Illinois EPA monitors and evaluates water quality. The Illinois EPA's 2024 Integrated Water Quality Report explains that measures such as dissolved oxygen and pH are used to help assess Aquatic Life Use in streams, and the agency reports that it has sampled about 3,500 stream stations statewide. Together, these studies and reports provide a useful benchmark for comparing the spatial and seasonal patterns found in this project.

Project Timeline

- Week 1: Finalize the refined research question and complete data preprocessing for the four primary indicators.
- Week 2: Produce exploratory summaries and visualizations for seasonal and site-level variation.
- Week 3: Apply comparison and data mining methods such as clustering or correlation analysis.
- Week 4: Interpret results, compare them with existing findings, and prepare the final report and presentation.

Sources Mentioned

Read, E. K., Carr, L., DeCicco, L., Dugan, H. A., Hanson, P. C., Hart, J. A., Kreft, J., Read, J. S., & Winslow, L. A. (2017). Water quality data for national-scale aquatic research: The Water Quality Portal. *Water Resources Research*, 53(2), 1735–1745.
<https://doi.org/10.1002/2016WR019993>

Mohebbi, A., & Akbariyeh, S. (2024). Non-parametric analysis of nitrogen trends in the form of nitrate and nitrite in rivers and streams of the contiguous United States for 1990–2019. *International Journal of River Basin Management*, 22(2), 129–141.
<https://doi.org/10.1080/15715124.2022.2114484>

Illinois Environmental Protection Agency. (2024). Illinois Integrated Water Quality Report and Section 303(d) List. <https://epa.illinois.gov/content/dam/soi/en/web/epa/topics/water-quality/watershed-management/tmdls/documents/2024-ir/2024%20Integrated%20Report%20and%20303d%20list2.pdf>

Illinois Environmental Protection Agency. (n.d.). River and stream monitoring.
<https://epa.illinois.gov/topics/water-quality/monitoring/river-and-stream.html>